

Music-Gen v4 — Cycles 50-52

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Music-Gen v4 — Cycles 50-52

Abstract

Cycles 50-52 extended the preservation-only heartbeat cadence into its ninth, tenth, and eleventh consecutive substantive-heartbeat cycles and opened a fourth operator-authority surface — the POR consolidation strategy proposal that the prior range’s audit had recommended — while all six previously-standing operator-authority escalations remained preserved verbatim with `blocked_on_operator=true`. Cycle 50 authored the POR consolidation strategy proposal at `docs/v4_por_consolidation_strategy_proposal_c40.md` (SHA `8cffc1cecf8fed87...`) presenting three named options (`OPT_1` / `OPT_2` / `OPT_3`) for operator selection, parallel to the `c31` / `c34` shadow-zone hygiene precedent, without pre-adjudicating between them; the document is a proposal awaiting operator selection, not a wait-on-operator memo (the anti-stall rule ban applies to memos that halt substantive work on operator response — this proposal is a substantive deliverable that lands independent of the operator’s eventual choice). Cycle 51 continued the preservation cadence but deviated from the `c35-c40` file-naming convention on the Priority 1 chain-supersede event (using `c41-long-exposure-absent-preservation.json` rather than the established `c<N>-emitter-writer-boundary-preservation.json` pattern); the deviation was flagged by the `c41` audit’s D-2 and codified in the `c42` brief’s M-1 as “`c35-c40` pattern is canonical.” Cycle 52 executed against the codified M-1 naming (`c42-emitter-writer-boundary-preservation.json`) while pointing `supersedes_path` at `c51`’s actual on-disk filename per invariant (d) — a reverse-case cleanup of the `c51` deviation that preserves chain-integrity via a string pointer to the real predecessor rather than the canonical-name predecessor. The chain-supersede lineage on the workspace-forced `OPT_B` ledger-emitter exemption now runs eight consecutive cycles `c34` → `c41` → `c42` (eight preservation events past the original `c34` fork); the stand-pat lineage on the POR shadow-drift strengthening question now runs nine consecutive cycles `c34` → `c42`. Cycle 52 also honestly disclosed a POR shadow-zone off-by-one — the brief expected 858 parseable Milestones at `c42` open (assuming `c41` registered 13 rows: `845 + 13`); on-disk was 859 (`c41` actually registered 14 rows per the `c41` audit’s own `+14` delta note); delta zero versus `c41` close baseline; per invariant (d) + FD-1 on-disk-authoritative this is not a shadow-zone breach but a brief-expectation off-by-one, cross-checkable via `python3 tools/_c32_por_count.py`. All six operator escalations remain preserved verbatim with correct `carried_from`

cycle bumps per c42 brief calibration (non-CG bass 13, metric-semantics 13, drums-fine 13, v2-bass-fine 12, guitar-fine 12, composite-FP-drift 11); all blocked_on_operator=true; _manager/exhaustive at six. Test coverage grew in-place from 28 → 30 → 32 → 34 cases on tests/test_c30_legacy_mode_regression.py across the range with the standalone eight-case OP-1 serial-lock suite unchanged; cross-cycle total 42/42 PASS at range close. Ledger grew 1636 → 1649 → 1663 across the range; POR parseable-Milestones traced 832 → 845 → 859 → 873 (four data points now in the +12/+14/+14 heartbeat band, structurally attributable per c34 empirical proof to the four Track B/C/D honest-deferral rows plus housekeeping-tail plus preservation events plus test-extension rows). Nine read-only anchors byte-identical pre-vs-post at range close including the new c40 consolidation strategy proposal document. Independent audit returned **VALIDATED** with three informational disclosures (two correct per invariant (d), one minor brief-reading nit that recommends future workers read a brief's "on-disk is X; NOT Y" pattern as a preemptive flag rather than a re-committed error). Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Canonical 7-key env_pin_sha256=2ac444c3...922ca unchanged; FD-16(a) re-issue not triggered. M-V4-SHOWCASE-1 status unchanged (LANDS_pending_operator). Forward campaign momentum remains operator-authority-owned across all four blocked surfaces (composite-FP-drift adjudication; consolidation-proposal selection; non-CG bass acceptance policy; metric-semantics adjudication).

Introduction

The Music-Gen v4 closure campaign is driving through seven strictly-ordered milestones toward a clean close. Cycles 50-52 are the ninth, tenth, and eleventh consecutive substantive-heartbeat cycles under the c36 auditor's M-1 / M-2 terminal contracts. All substantive stage-2 sweep work remains gated on operator adjudication that has not arrived through live_guidance. The prior range's audit had recommended that a subsequent cycle propose a POR consolidation strategy parallel to earlier shadow-zone hygiene precedent if the operator-adjudication wait continued; that recommendation was executed at Cycle 50 as a first-class substantive deliverable, opening what became Priority 8 in the subsequent cycles' briefs — operator selection among three named options presented in the proposal document.

Two continuing side-question chains from earlier ranges continue to shape each cycle: the workspace-forced OPT_B ledger-emitter exemption (Priority 1 chain-supersede on each cycle's re-probe of long_exposure/, ABSENT throughout the range) and the POR shadow-drift strengthening question (Priority 2 stand-pat on each cycle's scan of live_guidance for an operator-supplied c31/c32 POR snapshot, absent throughout the range). Both are terminal-contract patterns; the c14 string-typed supersedes_path lemma is exercised each cycle rather than treated as dormant convention.

Approach

Each cycle executed the same nine-priority template (extended from the prior range's eight-priority template by the addition of Priority 8 for the consolidation-proposal question):

- **Priority 0 (preserve six escalations).** Verify all six escalation sidecars under data/v4/_manager/ byte-identical versus the prior cycle's attestation SHAs; bump carried_from_cycle on rollup per calibration. No memo mutated; no seventh escalation opened.
- **Priority 1 (chain-continuation re-probe).** Execute test -d long_exposure, confirm ABSENT, emit chain-supersede preservation event with string supersedes_path per the c14 lemma pointing at the prior cycle's actual on-disk filename.
- **Priority 2 (POR stand-pat continuation).** Scan live_guidance for operator-supplied c31/c32 POR snapshot, confirm absent, emit stand-pat preservation event chain-superseding the prior cycle's stand-pat via string supersedes_path.

- **Priority 3 (composite-FP-drift operator adjudication).** Scan for PATH_A / PATH_B / PATH_C adjudication token, confirm absent, correctly skip.
- **Priority 4 (Peach Dream stem manifest anchor).** Verify SHA c4944ee80dfe446b... byte-identical; carry invariant-(d) divergence disclosure on the deferral row; no separate sidecar opened.
- **Priority 5 (honest-deferral rows for Track B/C/D).** Emit four rows with concrete resume commands (Disco A bass stage-2; Rome bass stage-2; Peach Dream bass stage-2 with invariant-(d) disclosure; WIG + Disco A drums stage-1).
- **Priority 6 (POR shadow-zone hold verification).** Verify parseable_milestones at cycle open matches prior cycle close baseline (delta zero); post-registration count grows by exactly the number of ledger rows emitted this cycle.
- **Priority 7 (Track F test-suite extension).** Extend tests/test_c30_legacy_mode_regression.py in-place by two cases per cycle capturing the cycle's new preservation and stand-pat events' shape and chain-integrity.
- **Priority 8 (POR consolidation strategy proposal — new this range).** At Cycle 50, author the proposal document with three named options for operator selection. In Cycles 51 and 52, verify the doc SHA byte-identical (8cffc1cecf8fed87...); scan live_guidance for OPT_1 / OPT_2 / OPT_3 selection token; skip if absent.

Cycle 50 specific. Authored docs/v4_por_consolidation_strategy_proposal_c40.md (SHA 8cffc1cecf8fed87...) with three named consolidation options presenting the trade-space to the operator without pre-adjudicating: OPT_1 (aggressive compaction that folds the accumulated Track B/C/D deferral-row set into a stable summary; single high-cost cycle), OPT_2 (progressive compaction across multiple cycles), OPT_3 (retain current mechanical pattern; explicit no-op). The proposal parallels the c31 / c34 shadow-zone hygiene precedent as a substantive deliverable that lands independent of the operator's eventual selection.

Cycle 51 specific. Deviated from the c35-c40 file-naming convention on the Priority 1 preservation event by using c41-long-exposure-absent-preservation.json rather than c<N>-emitter-writer-boundary-preservation.json. The c41 audit's D-2 flagged the deviation; the subsequent c42 brief's M-1 codified the c35-c40 pattern as canonical and directed subsequent cycles back to it.

Cycle 52 specific. Executed the c42 brief's M-1 codified naming (c42-emitter-writer-boundary-preservation.json) while pointing supersedes_path at Cycle 51's actual on-disk filename (c41-long-exposure-absent-preservation.json) per invariant (d). This is the reverse case of the c41 deviation: it restores the canonical naming going forward while preserving chain-integrity via a string pointer to the real predecessor rather than to a canonical-name that does not exist on disk. Also disclosed a POR shadow-zone off-by-one: brief expected 858 parseable Milestones at cycle open (assuming Cycle 51 registered 13 rows: 845 + 13), on-disk was 859 (Cycle 51 actually registered 14 rows per its own audit's +14 delta note); delta zero versus Cycle 51 close; per invariant (d) + FD-1 on-disk-authoritative this is not a shadow-zone breach.

Discipline guards asserted across the range. All AST-scannable invariants pass: no PRNG imports (random.*, np.random.*), no sidecar_nonfactor, no VST3 state APIs, no --verify-det bypass, /usr/bin/python3 interpreter guard implicit via the c34+ pattern. supersedes_path typed as string (never list) throughout, exercised across six new preservation events plus one new-attestation shadow-zone-hold with supersedes_path: null in Cycle 52 per c14 lemma. Canonical 7-key env_pin_sha256=2ac44c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842a922ca unchanged; FD-16(a) re-issue not triggered. FD-16(c) replay-proof invariant carried; no new renders in the range. OP-1 serial-lock invariant remains in force. Zero SF2_CONFIRMED verdicts on non-CG bass. No wait-on-operator memo emitted.

Findings

POR consolidation strategy proposal landed as first-class substantive deliverable

docs/v4_por_consolidation_strategy_proposal_c40.md (SHA 8cffc1cecf8fed87...) landed at Cycle 50 in response to the prior range's audit M-1 recommendation. The document presents three named options for operator selection parallel to the c31 / c34 shadow-zone hygiene precedent, opens Priority 8 in subsequent cycles' briefs, and holds the SHA byte-identical across Cycles 51 and 52. This is a substantive deliverable — the document itself is landed independent of the operator's eventual selection — that opens a fourth operator-authority surface without violating the anti-stall rule: the ban on wait-on-operator memos applies to memos that halt substantive work on operator response, and this document is itself the substantive work.

Chain-supersede lineage matured to eight cycles; stand-pat lineage to nine

The workspace-forced OPT_B ledger-emitter exemption chain now runs eight consecutive cycles past the c34 fork: c34 fork → c35 → c36 → c37 → c38 → c39 → c40 → c41 → c42. The POR shadow-drift stand-pat chain now runs nine consecutive cycles past the c34 empirical proof: c34 empirical proof → c35 blocker → c36 stand-pat → c37 → c38 → c39 → c40 → c41 → c42. Each cycle's `supersedes_path` is a string pointing at the immediately-prior cycle's actual on-disk event path per the c14 lemma; the range's Cycle 52 reverse-case cleanup (pointing at Cycle 51's off-canonical-name file) demonstrates the lemma's on-disk-authoritative discipline in action.

Six operator escalations preserved verbatim with `carried_from_cycle` bumps

All six escalation sidecars under `data/v4/_manager/` verified byte-identical at Cycle 52 close with correct `carried_from_cycle` calibration per c42 brief:

Escalation	Origin	<code>carried_from_cycle</code> at c42
SHOWCASE-1-non-cg-bass-acceptance-policy	c7	13
M-V4-METRIC-SEMANTICS-c16	c16	13
CERT-fine-fit-sf2-drums-legacy-halt	c30	13
CERT-fine-fit-sf2-v2-legacy-halt	c31	12
CERT-fine-fit-sf2-guitar-legacy-halt	c31	12
CERT-composite-fp-drift-adjudication-c32	c32	11

The c32 composite-FP-drift consolidation memo continues to link but not close the three per-driver fine-fit HALT predecessors. All six carry correct `carried_from_cycle` semantics; all `blocked_on_operator=true`; all discoverable via both `grep _manager/M-V4-CERT` and `ls data/v4/_manager/`. `_manager/` remains exhaustive at six — no seventh escalation opened. `SF2_CONFIRMED` remains FORBIDDEN on non-CG bass.

Three honest disclosures per invariant (d) at Cycle 52 audit

- **D-1 POR shadow-zone off-by-one.** Brief expected 858 parseable Milestones at c42 open assuming Cycle 51 registered 13 rows; on-disk was 859 (Cycle 51 actually registered 14 rows per its own audit's +14 delta note). Delta zero versus Cycle 51 close baseline. Per invariant (d) + FD-1 on-disk-authoritative this is not a shadow-zone breach but a brief-expectation off-by-one; cross-checkable via `python3 tools/_c32_por_count.py` returns 859. Consistent class with the c41 audit's D-1.

- **D-2 Priority 1 naming-convention adoption per brief M-1 codification.** Brief M-1 codified “c34-c40 use c<N>-emitter-writer-boundary-preservation.json” but Cycle 51 deviated to c41-long-exposure-absent-preservation.json. Cycle 52 adopted the M-1 canonical naming (c42-emitter-writer-boundary-preservation.json) with `supersedes_path` string-typed → Cycle 51’s actual on-disk filename per FD-1 + invariant (d). Reverse case of Cycle 51’s deviation. Chain-integrity intact.
- **D-3 Consolidation-doc SHA cross-reference (minor brief-reading nit).** The Cycle 52 worker’s D-3 disclosure treated a brief cross-reference to 29a1610b... for the consolidation proposal as a recurrence of the c41 audit’s D-2 error; in fact the brief itself preemptively flagged the class (“on-disk is 8cffc1cecf8fed87...; NOT 29a1610b... which per c41 audit D-2 was the invariants doc, not the proposal doc”) and did not re-commit the error. The disclosure is superfluous but not harmful; both docs verified byte-identical pre-vs-post. Recommendation: future workers should read a brief’s “on-disk is X; NOT Y” pattern as a preemptive flag rather than a re-committed error.

Read-only anchors held; test coverage grew 28 → 34 in-place

Nine read-only anchors verified byte-identical pre-vs-post at range close: `scripts/sound_match/objective.py` 8087ce80...; `scripts/sound_match/_sweep_hygiene_c27.py` 771ff42b...; `scripts/sound_match/_serial_lock_op1.py` 121809db...; `docs/agent_picks_selection_invariants.md` 29a1610b...; `docs/emitter_exemption_policy.md` fd2c33a7...; `docs/v4_por_consolidation_strategy_proposal_c40.md` 8cffc1c... (new this range); `data/v4/diagnostics/c34_por_delta_proof.json` 3b0e4d95...; `Peach Dream stem manifest` c4944ee8...; `data/v4/deliveries/31a164f845f8e27e/cg_ab_mix.wav` 6e13e007...

Cross-cycle regression coverage on `tests/test_c30_legacy_mode_regression.py` extended in-place by two cases per cycle across the range: c40 30 → c41 32 → c42 34. Standalone `tests/test_fine_fit_serial_lock_c32.py` unchanged at 8/8. Cross-cycle total 42/42 PASS at range close.

POR arithmetic tracked +12 / +14 / +14 across the range

Parseable-Milestones trajectory across the range: 832 (post-c39) → 845 (post-c50) → 859 (post-c51) → 873 (post-c52). Each cycle’s post-registration count matches the prior cycle’s close-baseline plus that cycle’s emitted ledger-row count exactly (13 for Cycle 50; 14 for Cycles 51 and 52). Ledger grew 1636 → 1649 → 1663 across the range. POR baseline 745 (c31 counting-method drift baseline) maintained throughout.

Audit outcome

VALIDATED at range close against all nine validators (a)–(i) of the c40 / c41 expanded validator matrix (extended from the prior range’s eight-validator matrix by the addition of (i) test-suite growth cadence). Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Three informational disclosures per invariant (d): D-1 (POR off-by-one, correct per FD-1 on-disk-authoritative), D-2 (naming-convention adoption, correct reverse-case cleanup), D-3 (minor brief-reading nit on the D-2/D-3 cross-reference pattern).

Discussion

Three things about this range are worth naming.

First, the range demonstrates that adding a new operator-authority surface (Priority 8 consolidation-proposal selection) does not compromise the discipline of the preservation-only cadence. The proposal document is landed as a first-class substantive deliverable at Cycle 50, and Cycles 51 and 52 verify its SHA byte-identical, scan `live_guidance` for a selection

token, and skip cleanly when absent — exactly the pattern already established for Priorities 3 (composite-FP-drift adjudication) and 0 (six standing escalations). The anti-stall rule ban on wait-on-operator memos applies to memos that halt substantive work; this proposal is itself substantive work that lands independently, and the priority tracking it in subsequent cycles is a preserve-and-skip discipline, not a wait-on-operator posture. This is the correct shape for opening a new operator-authority surface under the campaign’s discipline invariants: land the substantive artifact, add a preservation-and-scan priority for subsequent cycles, and continue.

Second, the Cycle 51 → Cycle 52 naming-convention deviation-and-cleanup demonstrates the c14 string-typed `supersedes_path` lemma working as designed under real drift. Cycle 51’s file-naming deviation was a small procedural slip (`c41-long-exposure-absent-preservation.json` rather than `c41-emitter-writer-boundary-preservation.json`); the c41 auditor flagged it; the c42 brief’s M-1 codified the canonical pattern; the Cycle 52 worker executed the codified naming going forward while pointing `supersedes_path` at the actual on-disk predecessor. Two properties survive intact: the canonical-name pattern is restored for future cycles (so the codified naming holds), and the chain-integrity is preserved via a string pointer to the actual on-disk filename per invariant (d) + FD-1 on-disk-authoritative (so no future audit finds a chain break). This is what a discipline lemma looks like when it is genuinely load-bearing: it survives drift without requiring anchor mutation.

Third, the +12/+14/+14 POR arithmetic pattern across the range confirms the c34 empirical proof attribution even under a small emitted-row variation. The c34 proof attributed the accumulating shadow-zone growth to a four-row Track B/C/D deferral set plus the house-keeping tail plus preservation events plus test-extension rows. Cycle 50 emitted 13 rows (the +1 versus the earlier +12 pattern is the new Priority 8 rollup event registering the consolidation proposal). Cycles 51 and 52 emitted 14 rows each (an additional +1 attributable to the standing Priority 8 preservation-and-scan pattern). Each variation is fully accounted-for at row-level; the attribution model is not broken by the new priority; the operator-supplied c31/c32 POR snapshot would still fire the Priority 2 close event if it arrived. This continues to support the informational-not-blocking characterization of the POR growth pattern.

Open questions

- **Composite-FP-drift operator adjudication.** Three named paths (A accept render-level bar via new invariant (f); B hold strict; C harden `objective.py`). Path A remains the single-action resolution across all three fine-fit drivers. No adjudication in `live_guidance` across the range.
- **Consolidation-proposal operator selection (new this range).** Three named options (OPT_1 aggressive compaction; OPT_2 progressive compaction; OPT_3 retain current pattern) in `docs/v4_por_consolidation_strategy_proposal_c40.md`. No selection token in `live_guidance` across the range.
- **Non-CG bass acceptance-policy escalation.** Remains `blocked_on_operator=true`. SF2_CONFIRMED FORBIDDEN. Systematic 4-arc composite-vs-source-of-truth finding still predicts OPT2 (refuse extension + OPT3 htdemucs bass fallback) as the invariant-compliant outcome.
- **Metric-semantics escalation.** Remains `blocked_on_operator=true` with `carried_from_cycle=13`.
- **Emitter writer-boundary chain.** Eight-cycle chain-supersede continues if `long_exposure/` remains ABSENT. Canonical naming pattern restored at c42 per M-1 codification.
- **POR shadow-drift stand-pat.** Nine-cycle chain continues absent operator-supplied c31/c32 POR snapshot. c34 empirical proof transitively holds.

- **POR consolidation-proposal preservation-and-scan.** Subsequent cycles verify the c40 proposal doc SHA byte-identical and scan live_guidance for OPT_1 / OPT_2 / OPT_3 selection token; execute the chosen option when it arrives, otherwise skip.
- **Expected next-cycle open state.** parseable_milestones = 873; six escalation memo SHAs unchanged; consolidation proposal SHA unchanged at 8cffc1c...; long_exposure/ ABSENT; c42 preservation + stand-pat events on disk byte-identical; 42/42 test suite green pre-extension; chain-length P1 = 9, P2 = 10 at next-cycle close; 44/44 test target.
- **Four operator triggers that would change the state.** (a) A c31/c32 POR snapshot for Priority 2 close; (b) a PATH_A / PATH_B / PATH_C adjudication token for the composite-FP-drift memo; (c) an OPT_1 / OPT_2 / OPT_3 selection token for the consolidation proposal; (d) resolution of the non-CG bass acceptance policy. All four remain absent.

Appendix: Provenance

Directive. Execute the Music-Gen v4 closure campaign; pursue milestones in strict order starting with M-V4-CERT-1 and M-V4-PROFILES-1; drive to a clean close.

Cycle range. cycles 50-52.

Working directory. /home/user/long-exposure-runs/music-gen.

Session references.

- Cycle 50 researcher a0d318ef-f459-4c1b-bdab-4b4ab14b5fd5; worker ff7328d1-5164-495f-a453-a3d7e4361ba7; auditor f4dc3a1a-aef3-439a-8edc-3985792f49c7.
- Cycle 51 researcher 1905faa8-3eed-4586-b88e-f59e5e663735; worker 19ed284b-5f3d-45f3-b00e-bef719b7a4dc; auditor 68e1d00e-fa17-4deb-b442-96de2bcf769d.
- Cycle 52 researcher daeb9e01-9eac-492b-a7c0-d0f2382b51e4; worker d63c7c8e-4fdd-481e-bfc2-f361b093764e; auditor 909c218a-9803-4eb0-9a55-b20403ec4034.

Audit verdict. VALIDATED at range close against all nine validators (a)-(i). Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Three informational disclosures per invariant (d): D-1 POR off-by-one (correct per FD-1 on-disk-authoritative); D-2 Priority 1 naming-convention adoption per brief M-1 codification (correct reverse-case cleanup); D-3 minor brief-reading nit on the D-2/D-3 cross-reference pattern.

Terminal deliverables landed this range.

- docs/v4_por_consolidation_strategy_proposal_c40.md (SHA 8cffc1cecf8fed87...) — first-class substantive deliverable presenting three named options (OPT_1 / OPT_2 / OPT_3) for operator selection parallel to c31 / c34 shadow-zone hygiene precedent; opens Priority 8 in subsequent cycles.
- Three chain-supersede preservation events on the workspace-forced OPT_B ledger-emitter exemption (data/v4/_selection/c{40,41,42}-*-preservation.json), each string-typed supersedes_path → prior-cycle actual on-disk filename per c14 lemma. Cycle 51's file naming deviated (c41-long-exposure-absent-preservation.json); Cycle 52 restored canonical naming per brief M-1 codification.
- Three stand-pat preservation events on the POR shadow-drift strengthening question (data/v4/_selection/c{40,41,42}-por-drift-preservation.json), each string-typed supersedes_path → prior-cycle stand-pat.
- Twelve honest-deferral rows total across the range for Track B/C/D (four per cycle) with concrete resume commands.
- Six new test cases across the range (two per cycle) extending tests/test_c30_legacy_mode_regression.py in-place 28 → 30 → 32 → 34; standalone tests/test_fine_fit_serial_lock_c32.py unchanged at 8/8; cross-cycle total 42/42 PASS at range close.

- Three housekeeping-triad sequences (one per cycle): `_run/cycle_N_closed` → `_archive/cycle-N-scratch/` → `_infra/adopt-cycleN-tests`.

Ledger + POR arithmetic across the range.

- Cycle 50: ledger +13 (Priority 8 rollup adds one row); POR 832 → 845.
- Cycle 51: ledger +14 (Priority 8 preservation-and-scan adds one row); POR 845 → 859.
- Cycle 52: ledger +14; POR 859 → 873.
- Cumulative range: ledger 1636 → 1663; POR 832 → 873. Each cycle open matches prior cycle close (delta zero). POR baseline 745 (c31 counting-method drift baseline) maintained.

Read-only anchors preserved byte-identical pre-vs-post (9 verified at range close).

- `scripts/sound_match/objective.py` 8087ce80...
- `scripts/sound_match/_sweep_hygiene_c27.py` 771ff42b...
- `scripts/sound_match/_serial_lock_op1.py` 121809db...
- `docs/agent_picks_selection_invariants.md` 29a1610b...
- `docs/emitter_exemption_policy.md` fd2c33a7...
- `docs/v4_por_consolidation_strategy_proposal_c40.md` 8cffc1c... (new this range)
- `data/v4/diagnostics/c34_por_delta_proof.json` 3b0e4d95...
- Peach Dream stem manifest c4944ee8...
- `data/v4/deliveries/31a164f845f8e27e/cg_ab_mix.wav` 6e13e007...

Chain-supersede lineage at range close.

- Priority 1 (emitter-writer-boundary): c34 fork → c35 → c36 → c37 → c38 → c39 → c40 → c41 → c42. Chain-length 8.
- Priority 2 (POR shadow-drift stand-pat): c34 empirical proof → c35 blocker → c36 → c37 → c38 → c39 → c40 → c41 → c42. Chain-length 9.

Six operator escalations preserved verbatim with correct `carried_from_cycle` bumps (non-CG bass 13, metric-semantics 13, drums-fine 13, v2-bass-fine 12, guitar-fine 12, composite-FP-drift 11). All `blocked_on_operator=true`. `_manager/` exhaustive at six.

Substantive-heartbeat streak at range close. c33 → c42 = 11 consecutive substantive-heartbeat cycles under c36 auditor M-2 terminal contract.

Environment pin. Canonical 7-key `env_pin_sha256=2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842` unchanged this range; FD-16(a) re-issue not triggered.

Discipline guards asserted (AST-scannable). No PRNG imports, no `sidecar_nonfactor`, no VST3 state APIs, no `--verify-det` bypass, `/usr/bin/python3` interpreter guard implicit via c34+ pattern. `supersedes_path` typed as string throughout (except one new-attestation P6 shadow-zone-hold at Cycle 52 with `supersedes_path: null`). Zero `SF2_CONFIRMED` verdicts on non-CG bass (invariant 9 FORBIDDEN). OP-1 serial-lock invariant in force. No wait-on-operator memo (banned per operator directive 2026-09-03 part 2); Priority 0 preserve-only posture satisfies genuine operator-authority carve-out.

Milestone status at range close.

- M-V4-CERT-1 — validated (E2E_DETERMINISM_HOLDS on the v3 spine).
- M-V4-PROFILES-1 CG (5/5 instruments) — validated.
- M-V4-PROFILES-1 non-CG bass — 2/4 `SF2_RULED_OUT` (Rome, Peach Dream); 2/4 `STILL_INDETERMINATE` (WIG, Disco A). Stage-2 re-work gated on operator adjudication.
- M-V4-PROFILES-1 non-CG drums — 0/4 (Track D unblocked at driver-regression level; stage-2 gated on operator adjudication).

- M-V4-PROFILES-1 non-CG guitar — 0/2 (WIG + Peach Dream guitar are NULL by earlier MIDI-probe).
- M-V4-SHOWCASE-1 CG — LANDS_pending_operator (cg_ab_mix.wav SHA 6e13e007... byte-identical since c17).
- M-V4-SHOWCASE-1 non-CG — BLOCKED on non-CG bass acceptance-policy escalation.
- M-V4-RULES-1 — scaffold landed c20; substantive implementation queued.
- M-V4-EAR-1 — not yet opened.
- M-V4-GEN-1 — conditional on M-V4-RULES + M-V4-EAR.
- M-V4-CLOSE-1 — c24 amendment landed; further amendments as substantive work completes.

Next-cycle first task. Continue the nine-priority preservation cadence per c42 auditor forward guidance: Priority 1 chain-continuation (re-probe long_exposure/; chain-supersede c42 preservation using the M-1 canonical naming c43-emitter-writer-boundary-preservation.json); Priority 2 stand-pat continuation; Priorities 3, 4, 5, 6, 7 as usual; Priority 8 preservation-and-scan (verify consolidation-proposal doc SHA byte-identical; scan live_guidance for OPT_1 / OPT_2 / OPT_3 selection token; skip if absent). Expected chain-lengths at next-cycle close: P1 = 9, P2 = 10. Expected test target: 44/44 PASS. Expected POR at next-cycle open: 873 (matches Cycle 52 close). If any of the four operator triggers arrives via live_guidance, execute the chosen path. Operator ear remains LANDS authority post-hoc per FD-6.